Guangyuan Weng

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RESEARCH INTERESTS

Multimodal AI; LLM evaluation and mechanistic interpretability; recommender systems; representation learning; spatiotemporal data mining; responsible AI.

EDUCATION

Northeastern University Sept. 2021 – May 2027 expected
Ph.D. Candidate, Computer Science; Advisor: Prof. Esteban Moro *Boston, MA*

ShanghaiTech University Sept. 2017 – July 2021
B.E., Computer Science and Technology; Thesis advisor: Prof. David J. Crandall (Indiana University) *Shanghai, China*

INDUSTRY EXPERIENCE

Chewy June – Aug. 2026
Research Scientist II Intern, MSO Data Science *Boston, MA*

- Engineered an end-to-end **multimodal GenAI pipeline** extracting structured visual and auditory semantics from video assets, integrating scene-aware keyframe sampling (FFmpeg), speech transcription (Whisper), and OCR into structured digests grounded to source timestamps.
- Developed an **LLM-as-a-judge evaluation framework** to benchmark candidate video-digest models; designed a human-in-the-loop system to discover, deduplicate, and calibrate high-dimensional creative attributes into an executable detection taxonomy.
- Formulated downstream **predictive models** (CTR and video completion) linking multimodal creative features to campaign KPIs, achieving a **15.9% lower prediction error (MAE)** over baselines across audience segments; deployed an interactive analytics dashboard presented to Chewy's **Chief Brand Officer and marketing leadership**.

Wormpex AI Research May – Aug. 2023
Research Intern; Advisor: Dr. Gang Hua (IEEE Fellow) *Bellevue, WA*

- Developed hybrid retrieval-augmented classification architectures for **long-tail visual recognition**, designing a **self-attention importance weighting module** in RAG to dynamically prioritize salient retrieved contexts and filter noise.
- Implemented and benchmarked models in **PyTorch**, building automated data ingestion, distributed training, and evaluation pipelines in close collaboration with research and engineering teams.

RESEARCH EXPERIENCE

Northeastern University, Social Urban Networks Lab Jan. 2024 – Present
Research Assistant; Advisor: Prof. Esteban Moro *Boston, MA*

LLM recommendation evaluation *Jan. 2026 – Present*

- Architected an end-to-end evaluation framework using counterfactual prompt perturbation, **dense vector embeddings**, and geospatial verification algorithms across **3 LLM families, 304 neighborhoods, and 5 U.S. metros**.
- Audited generative recommendation systems: identified a **36.8%** hallucination rate in open generation, and demonstrated that constrained retrieval eliminated fabrication while uncovering a **47.5%** systemic candidate coverage gap.

Recommender feedback simulation *June 2025 – Present*

- Developed a scalable **agent-based simulation of human–AI feedback loops**, evaluating how iterative **collaborative filtering with online retraining** shapes long-term user behavior, exposure fairness, and socioeconomic segregation.

Mobility representation learning *Jan. 2024 – June 2025*

- Developed a **mobility representation learning framework** trained on **25.4M trajectories across 11 U.S. metros**, mapping human mobility dynamics into latent metric spaces capturing behavioral connections beyond physical geography.
- Demonstrated that learned neural embeddings significantly outperform Euclidean distance in explaining urban mobility flows via gravity modeling (**Boston R^2: 0.61 vs. 0.33**), uncovering systemic socioeconomic and physical boundaries.

Northeastern University Jan. 2026 – Present
Project Leader, Mechanistic Interpretability; Advisor: Prof. David Bau *Boston, MA*

· Investigated internal representations in frontier open LLMs (**Llama-3.1-70B, Qwen-2.5-72B**) using **nnsight/NDIF** for residual-stream activation caching, linear probing, and **Representational Similarity Analysis (RSA)**.

· Discovered cultural asymmetries across **2,400 global cities**; developed **contrastive activation addition (activation steering)** to intervene on latent representations and align spatial reasoning behaviors.

Northeastern University, Visual Intelligence Lab Sept. 2021 – May 2023
Research Assistant; Advisor: Prof. Huaizu Jiang *Boston, MA*

· Investigated **vision-language models (VLMs)**, few-shot learning, and compositional reasoning; designed contrastive representation learning objectives on synthetic image-graph pairs to improve transfer to real-world benchmarks.

Indiana University, Computer Vision Lab July 2020 – June 2021
Visiting Research Intern; Advisor: Prof. David J. Crandall *Bloomington, IN*

· Trained **Graph Convolutional Networks (GCNs)** to model action–object co-occurrence statistics in egocentric video; analyzed how visual priors impact out-of-distribution generalization in PyTorch.

ShanghaiTech University, Mobile Autonomous Robotic Systems Lab Sept. 2018 – Jan. 2020
Research Assistant; Advisor: Prof. Sören Schwertfeger *Shanghai, China*

· Built a mapping/**SLAM** robot on **ROS** with hardware-triggered **distributed sensor synchronization**; automated calibration and SLAM evaluation in C++/Python to produce high-resolution benchmark datasets.

SELECTED PUBLICATIONS AND MANUSCRIPTS

Recommendation Feedback Loops Shape Urban Income Segregation and Venue Visitation.
G. Weng, L. Chen, C. Beneduce, G. Mauro, M. Luca, T. Yabe, B. Lepri, L. Pappalardo, E. Moro. 2026. [Code]

Large language models create an uneven informational layer over cities.
L. Chen, **G. Weng**, E. Moro. 2026. [Code]

Mapping the Blind Spots: Cultural Asymmetries in LLM Geographic Representations.
G. Weng, A. Malik, G. Savcisens, T. Urmi, D. Bau. 2026.

Beyond Distance: Mobility Neural Embeddings Reveal Visible and Invisible Barriers in Urban Space.
G. Weng, M. Kim, Y.-Y. Ahn, E. Moro. *Science Advances* (accepted), 2026. [Code]

Action Recognition based on Cross-Situational Action-object Statistics.
S. Tsutsui, X. Wang, **G. Weng**, Y. Zhang, D. Crandall, C. Yu. *IEEE ICDL*, 2022.

Advanced mapping robot and high-resolution dataset.
H. Chen, Z. Yang, X. Zhao, **G. Weng**, et al. *Robotics and Autonomous Systems*, 2020.

Towards Generation and Evaluation of Comprehensive Mapping Robot Datasets.
H. Chen et al., including **G. Weng**. *ICRA Workshop on Dataset Generation and Benchmarking*, 2019.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, Rust, SQL, R, Shell/Bash.

Machine Learning & GenAI: PyTorch, Hugging Face, JAX, vLLM, LangChain, LangGraph, nnsight, scikit-learn, W&B.

Core CS & Systems: Algorithms & Data Structures, Distributed Systems, Scalable System Design, Automated Pipelines.

Cloud, Tools & Platforms: Linux/Unix, Git, Docker, Amazon Web Services (AWS), Google Cloud Platform (GCP), FFmpeg, ROS, Claude Code, Codex.

ADDITIONAL INFORMATION

Honors & Awards: Northeastern Outstanding PhD Award for Teaching (2026); PhD Network Travel Award (2025); IU International Research Fellowship (2020); ShanghaiTech Merit Student (Top 5%, 2020).

Selected Presentations: IC2S2 2025 Plenary Talk (top 1.5% of submissions); IC2S2 2024 Plenary Talk (top 2.8%).

Academic Service: Program Committee / Reviewer for NeurIPS (2026), IC2S2 (2024–2026), and NetSci (2025–2026).

Teaching Experience: Head TA: Foundations of AI (CS5100, Fall 2025), Mobile App Dev (CS5520, Spring/Fall 2024, Spring/Summer 2025). TA: Mobile App Dev (CS4520, Summer 2024), Pattern Recognition & Computer Vision (CS5330, Fall 2021/2022).